

## Lesson 5-7: Determine Surface Area of Prisms

Grade 6 Mathematics · Standard 6.GR.4

Name: \_\_\_\_\_ Date: \_\_\_\_\_

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

### Part A — Skills Check

Warm up with the lesson's core skills — one best answer each.

#### Question 1 SELECTED RESPONSE · 1 POINT

What is the surface area of a rectangular prism with  $l = 5$  cm,  $w = 4$  cm,  $h = 3$  cm?

- ☐ A 47 cm<sup>2</sup>
- ☐ B 60 cm<sup>2</sup>
- ☐ C 94 cm<sup>2</sup>
- ☐ D 94 cm<sup>3</sup>

#### Question 2 SELECTED RESPONSE · 1 POINT

How many faces does a triangular prism have?

- ☐ A 3
- ☐ B 4
- ☐ C 5
- ☐ D 6

#### Question 3 SELECTED RESPONSE · 1 POINT

A triangular prism has two triangular bases each with area 12 cm<sup>2</sup> and three rectangular faces with areas 30 cm<sup>2</sup>, 40 cm<sup>2</sup>, and 40 cm<sup>2</sup>. What is the total surface area?

- ☐ A 110 cm<sup>2</sup>
- ☐ B 122 cm<sup>2</sup>
- ☐ C 134 cm<sup>2</sup>
- ☐ D 134 cm<sup>3</sup>

**Question 4**    SELECTED RESPONSE · 1 POINT

A triangular prism has triangular bases with base 6 in and height 4 in. The prism is 10 in long and the two slant sides of the triangle are each 5 in. What is the surface area?

- ☐ A 124 in<sup>2</sup>
- ☐ B 160 in<sup>2</sup>
- ☐ C 184 in<sup>2</sup>
- ☐ D 184 in<sup>3</sup>

**Question 5**    SELECTED RESPONSE · 1 POINT

A rectangular time capsule box measures 3 in by 4 in by 5 in. How much wrapping paper covers its surface (no overlap)?

- ☐ A 47 in<sup>2</sup>
- ☐ B 60 in<sup>2</sup>
- ☐ C 94 in<sup>2</sup>
- ☐ D 94 in<sup>3</sup>

**Question 6**    SELECTED RESPONSE · 1 POINT

A rectangular capsule crate is 6 cm long, 5 cm wide, and 4 cm tall. How much cardboard is needed to cover all its faces?

- ☐ A 74 cm<sup>2</sup>
- ☐ B 120 cm<sup>2</sup>
- ☐ C 148 cm<sup>2</sup>
- ☐ D 148 cm<sup>3</sup>

**Part B — Test-Format Questions**

These match the state test's formats: two-part questions, select-all, and written responses.

**Question 7****TWO-PART QUESTION****Part A**

A rectangular prism measures 6 cm long, 4 cm wide, and 2 cm tall. What is its surface area?

*Fill in the bubble next to the one best answer.*

- ☐ A 24 square centimeters
- ☐ B 44 square centimeters
- ☐ C 48 square centimeters
- ☐ D 88 square centimeters

**Part B**

*Answer Part A before Part B — Part B asks about your reasoning.*

Why is surface area measured in square units while volume is measured in cubic units?

- ☐ A Surface area covers the outside faces, which are flat regions; volume fills the inside space.
- ☐ B Surface area is always smaller than volume.
- ☐ C Square units are used whenever a shape has straight sides.
- ☐ D Volume uses cubic units only for prisms.

**Question 8****WRITTEN RESPONSE · 2 POINTS****Type II Reasoning: Counting Only Three Faces**

Priya found the surface area of a box measuring 5 by 3 by 2 units. She wrote: ' $5 \times 3 + 5 \times 2 + 3 \times 2 = 15 + 10 + 6 = 31$  square units.'

Explain Priya's misconception and give the correct surface area.

*Write your answer in complete sentences. Show or explain your mathematical thinking.*

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## Part C — Show Your Thinking

Answer in writing, the way the state test's constructed responses work.

### Question 9 WRITTEN RESPONSE · 2 POINTS

Compare two closed containers: a rectangular prism ( $8 \times 6 \times 4$  in) and a triangular prism whose ends are right triangles with perpendicular legs 6 in and 8 in and hypotenuse 10 in. The triangular prism is 8 in long. Which uses less protective film to cover every face? Show your work.

*Write your answer in complete sentences. Show or explain your mathematical thinking.*

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When you finish, check your reasoning with your teacher or family. Your teacher has the scoring guide for the written responses.